REVIEW 1
OVERVIEW

This document includes the majority of the critiques made by Team 1. Some of them are irrelevant to our current context, others would need a lot more time to implement. I made sure to highlight the most relevant parts of the document with this

SUMMARY (TLDR):

LETTER OVERVIEW:

a. **Team 1 recommends the implementation of a façade design in order to simplify the "complicated structure". Most certainly not applicable.**
b. **Make sure every game has a "Return to Menu feature"**
c. **New use case for PlacingPiece in Tic-Tac-Toe (possibly in other games if missing?)**
d. **New use-case for ReturnHome button (check b)**
e. **Implement loading screen**

PLANNING DOCUMENT OVERVIEW:

a. ***Create separate project trackers for each subteam***
b. ***Avoid using general names. Be specific***
c. ***Include how many team members are working on which tasks***

PLANNING DOCUMENT OVERVIEW:

a. *Create separate project trackers for each subteam*
b. *Avoid using general names. Be specific*
c. *Include how many team members are working on which tasks*
d. *Apply task parallelism (already achieved)*

FEATURE PROPOSAL OVERVIEW:

a. *PROFILE PRIVACY AND FRIENDS FEATURES*
b. *PLAY AGAIN FEATURE*

CLASS DIAGRAM OVERVIEW:

a. *Very minor fix on (Backend)*
b. *Include missing methods in class diagram representation for ProfileCustomization (Frontend)*
c. *Make respective Leaderboard functions return an array with the top players instead of an int indicating user's current rank (Frontend)*
d. *Deal with extenuating circumstances (reconnects, disconnects, etc) in GameSessionHandler (Networking)*
e. *Add missing connections between lots of classes (Networking)*
    a. *Check DIAGRAMS / CLASS DIAGRAM / NETWORKING section*

USE CASE DIAGRAMS OVERVIEW

a. Backend
    a. **Change relationship from Validate Move to Check Game End to <<includes>>**
b. Frontend
        i. **Consider reviewing and adding <<Extends>> to:**
            1. **Landing Page -> All Connected**
            2. **Player Profile -> All Connected**
            3. **Customize Profile -> All Connected**
            4. **Choose Game -> All Connected**
            5. **Checkers Home -> All Connected**
            6. **Connect Four Home -> All Connected**
            7. **Tic Tac Toe Home -> All Connected**

c. Networking
        i. **Consider reviewing and adding <<Extends>> to:**
            1. **Launch Game-> Main Menu**
            2. **Network Manager-> Check Network, Verify Credentials**
            3. **Customize Profile -> All Connected**
            4. **Initialize Connection-> Connect Players**
            5. **Matchmaking Server -> Accept Match**
            6. **Accept Match -> Initiate Game**

USE-CASE DESCRIPTIONS:

- **Check Game End (Win/Lose/Tie)**
  - Recommended to change **Primary Actor** to **Player**
  - Add **Multiplayer Issues** to **Exceptions**
  - Change **Frequency of Use** to **Once Every Move is Made**
- **End Game**
  - Change **Primary Actor** to **System**
  - In **Postconditions,** clarify that Data is added to a player stats (i.e. Win record)
- **Start Game**
  - Possibly add to exceptions that if a Game is not available, Start Game cannot be performed
- **Choose a Side**
  - **Add extra channels to Channel to Actor.** Include Keyboard or Mouse Inputs, as this software is devised for more than touch-screen devices
- **Landing Page**
  - In Frequency of Use, consider using Low/Medium/High instead of "Multiple times" for most concise estimates.
- **Login**
  - **Maybe N/A given time constraints, but Team 1 suggested a "Remember Me" functionality and login time out in case the login fails multiple times**
- **View Leaderboard**
  - Change Frequency of Use to Medium/High. Such functionality is very commonly used.
  - For Secondary Actors, Team 1 proposed a change to "Database" or "Backend"
    - IMPORTANT: IF THIS IS DONE FOR VIEW LEADERBOARD, CONSIDER CHANGING EVERY OTHER INSTANCE WHERE "CUSTOMER SERVICE" is UTILIZED
- **View Other Player's Profiles**
  - **Postconditions** should be: "The player searched for stats are shown" or something similar
  - Change **Priority** to Low. This functionality is not essential for the system to run.
- **Profile Customization**
  - Change **Priority** to Low/Medium. Not essential for overall program running.
- **Customize Color Theme**
  - Change **Priority** to Low/Medium. Not essential for overall program running.
  - Change **Secondary Actor** to be the Database
- **Profile Image Selection**
  - Change **Priority** to Low/Medium. Not essential for overall program running.
  - Change **Secondary Actor** to be the Database
- **Set Profile Privacy Settings**
  - Change **Priority** to Low/Medium. Not essential for overall program running.
  - Change **Secondary Actor** to be the Database

- **Set Status Message**
  - Change **Priority** to Low/Medium. Not essential for overall program running.
  - Change **Secondary Actor** to be the Database
  - Consider adding an **Open Issue/Exception** that handles the size of "Status Message" (i.e. error if exceeds 300 characters)
- **Tic Tac Toe Home Menu**
  - Team 1 recommends that we account for cases where "a player wants to return back to choose game menu if it decides to not want to play"
- **No Match Found**
  - There seems to be two Use-Cases named No Match Found which seemingly tackle different functions
  - Team 1 recommends that they are turned into only one Use-Case
- **Game Instance**
  - **Rename it "Checkers Game Instance"**
- Another **No Match Found**
  - Rename the one that refers to the checker pieces to **"No Match Found Checkers"**
- **Checkers Leaderboard**
  - Specify the number of players to be displayed in the Leaderboard
- **View Game Stats (Aftermath)**
  - We do not specify how are good/bad moves determined. Possibly remove the sections that mention good/bad moves if we're not implementing anything of the sort
- **End game**
  - Change Network to System
  - In Scenario, include in Step2 that the system also checks for game end conditions. Make sure to properly explain this part (i.e. where does it go if a game end condition is met?)
- **Initialize Connection**
  - Change Primary Actor to Player. It is player that initiates the connection procedure
- **Accept Match**
  - Step3 and Step4 seem to be bad designs. If the player has a choice to accept a match based on the adversary rankings, they will always go for lower ranked adversaries, which implies that high-ranked players will hardly ever find matches**.**
- **Reject Match**
  - Refer to Accept Match (just above). Could still be implemented, but without displaying adversary rankings

Documents:

Letter:

1. *"Overall your class diagram was well designed. You had a great use of the observer design pattern for sending game data. There are some minor syntax error in the UML diagram which are marked in red, annotated and attached to this feedback deliverable.* ***We also recommend implementing a façade class to simplify the complicated structure****. With some recommended minor adjustments, the class diagram can be further improved."*
    a. **Team 1 recommends the implementation of a façade design in order to simplify the "complicated structure".**
    b. **Not applicable? Would imply considerable redesign of current code**

2. *"Firstly, the design is missing a way for player to return back from tic tac toe home to "choose game" menu if a player decides not to want to play tic tac toe anymore."*
    a. **This is directed towards the Tic-Tac-Toe game specifically, but we need to make sure every game implementation has a "Menu"/"Home" button that allows the user to return to Game Selection screen.**

3. *"There are few things that can be implemented to your game design that can further enhance your game design. An use case of placing pieces on the tic tac toe and a return option in tic tac toe home menu would be nice."*
    a. **One additional use case for PlacingPieceTicTacToe**
    b. **Also reiterating, one additional use case for ReturnHome**

4. *"For the tic tac toe game queue, we will recommend first searching the queue for valid opponent before adding the player into the queue."*
    a. **Team 1 recommends that opponent validation be done before enqueueing a selected player.**

5. *"We will also recommend making a separate use case or sequence diagram to get a better understanding of the process of game as it doesn't seem very clear."*
    a. **New use-case for "process of game"? Unclear feedback**

6. *"We will also recommend that you add a bit more emphasis on what the waiting process for the queue would be such as a loading screen icon or perhaps a prompt saying, "waiting for player". "*
    a. **Recommendation for a loading screen**

Planning Document:

1. *The project tracker excel file was not used. Please make sure you document and use the project tracker so the team stays on track and you can see the accountability of each team member.*
    a. **Weird feedback? I'm certain our team did use the project tracker**

2. *It may be beneficial for your teams to have overlapping tasks or goals running in parallel as it offers flexibility for your team to not bottleneck progression. For instance, it may be beneficial for your team to try to work on 2 games rather than finish 1 game completely then starting the next one.*
    a. ***Team 1 recommends parallelism of tasks in order to avoid bottleneck. This was clearly accomplished by our team in P3***

3. *We would recommend that you use project tracker for each team which would keep track of smaller and bigger milestones. Please make sure to label each milestones with details such as "Complete first game" should mention which first game is to be completed. You should also include how many team members of each sub team are working on which tasks.*
    a. ***Create separate project trackers for each subteam***
    b. ***Avoid using general names. Be specific***
    c. ***Include how many team members are working on which tasks***

FEATURE PROPOSAL

1. PROFILE PRIVACY SYSTEM

a. *To address players' privacy preferences and enhance the social experience, this proposal introduces **a friends feature** and **privacy controls,** allowing players to manage who can view their profiles. Players will be able to select options to make their profiles visible to everyone, only friends, or completely private. Also, this update provides players with a customized experience, enabling them to control the visibility of specific profile sections, such as game stats, achievements, and play history.*
    i. Implement a friends system
    ii. Implement privacy controls that allow the user to limit the access to certain parts or the entire profile in three types of access: **everyone, only friends, private**
b. For more info, please refer on Team 1's solution, refer to Expected Impacts and Design Suggestions
c. These imply drastic design changes, most assuredly not applicable.

2. POST-GAME "PLAY AGAIN" FEATURE

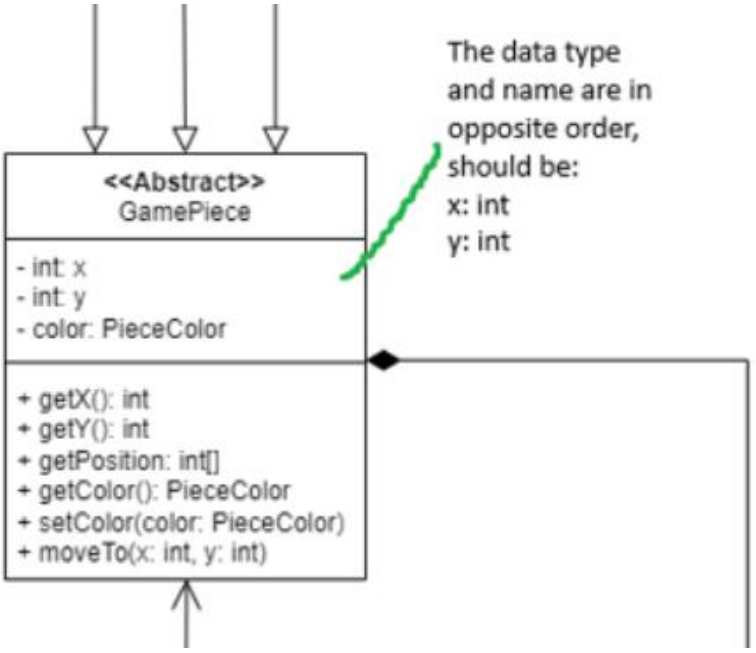
a. *Implementation of "Play Again" button/screen to employ friendly competition and rivalry, by allowing users to compete in a game once again in the post-game menu, allowing them tally wins against each other and to measure whomever has the natural competitive advantage. This screen could also display the other player's username. This would contribute to the community building aspect of the application, as recommended by the project document.*
    i. ***Create a "Play Again" feature that allows players to send a "Play Again Request" to the other player after a games has ended***
    ii. ***Possibly viable? Depends on Project status.***

Diagrams:

CLASS DIAGRAM:

Backend:

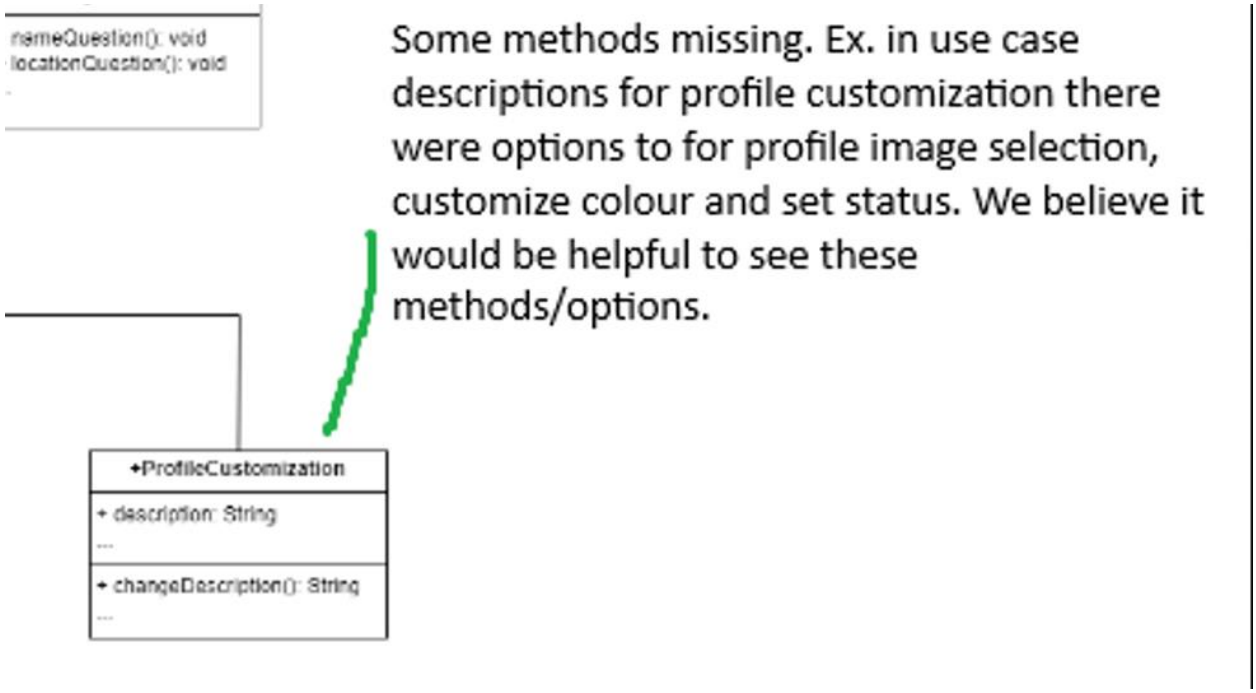
1.



a. Minor fix. In <<Abstract>> GamePiece, change "int: x" to "x: int"

Frontend:

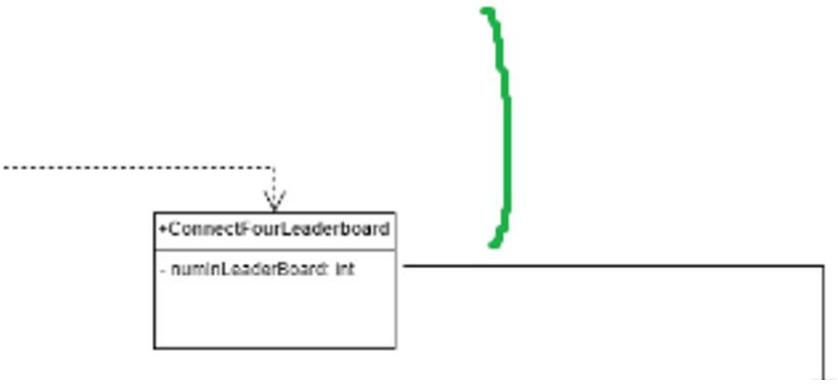
1.



a. **Are the features above for profile customization still part of the frontend design?**
    a. **PFP selection, customize color and set status**
b. **If so, Team 1 recommends including some of these methods in the ProfileCustomization class representation**

2.

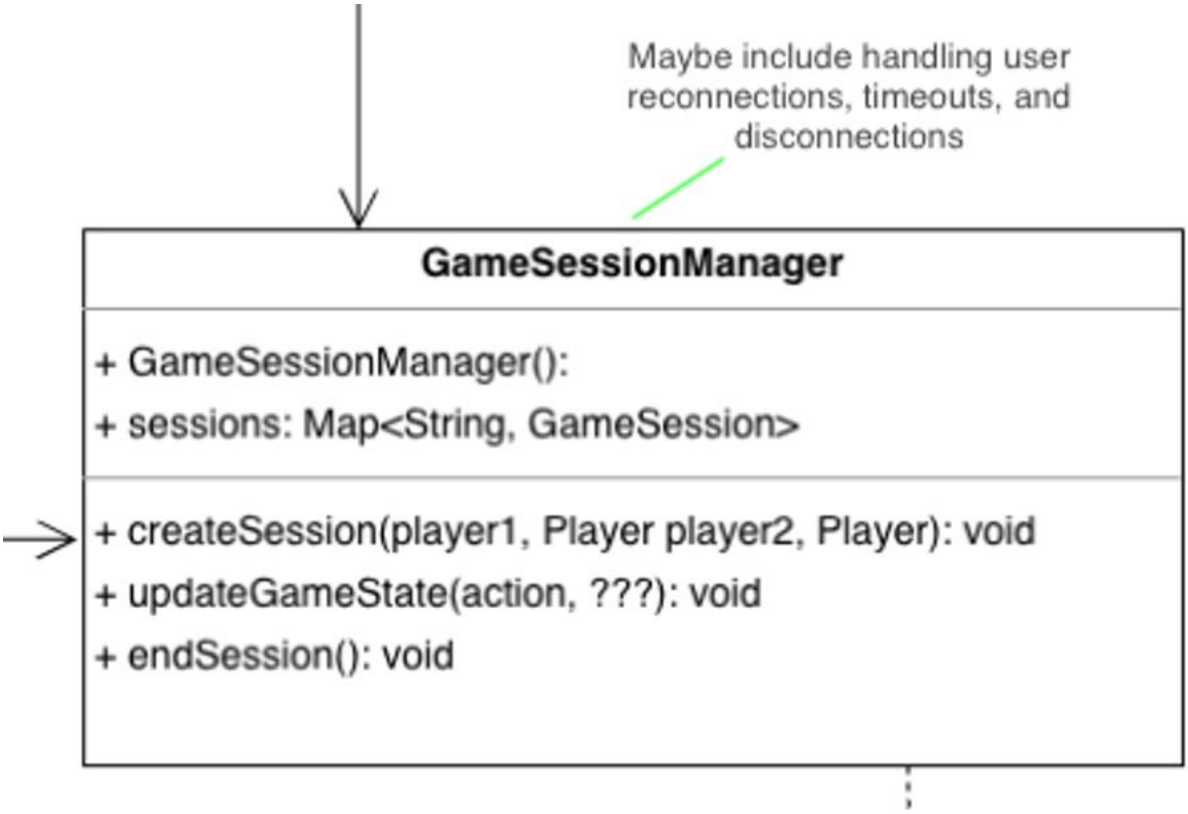
For leaderboard it looks like it just returns your rank (an int). I would consider implemented a leaderboard that shows the top players not just returning an int of the players rank.



a. **Team 1 recommends a more complex implementation of Leaderboard**
b. **Instead of returning the player's rank, it should instead return a list of the top players**

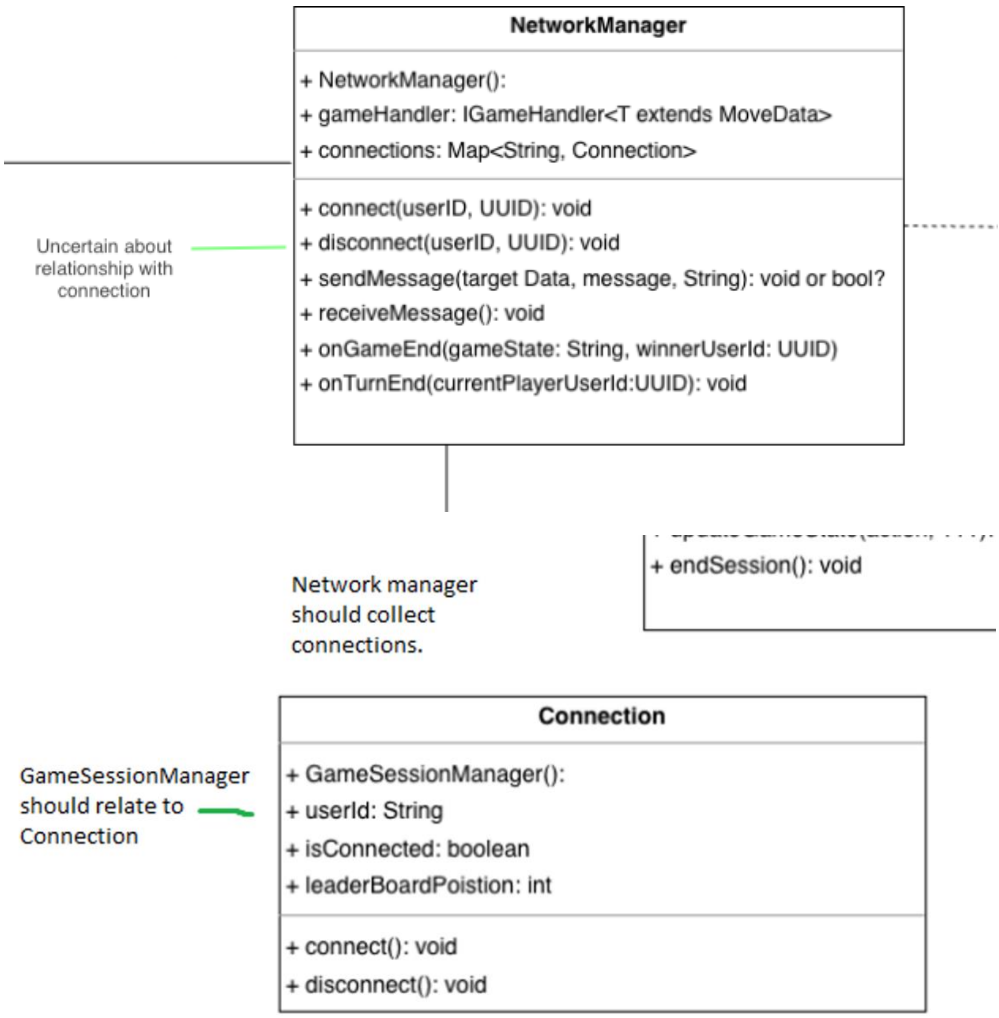
Networking:

1.



a. **In GameSessionManager, have Handlers to deal with reconnections, timeouts and disconnections**

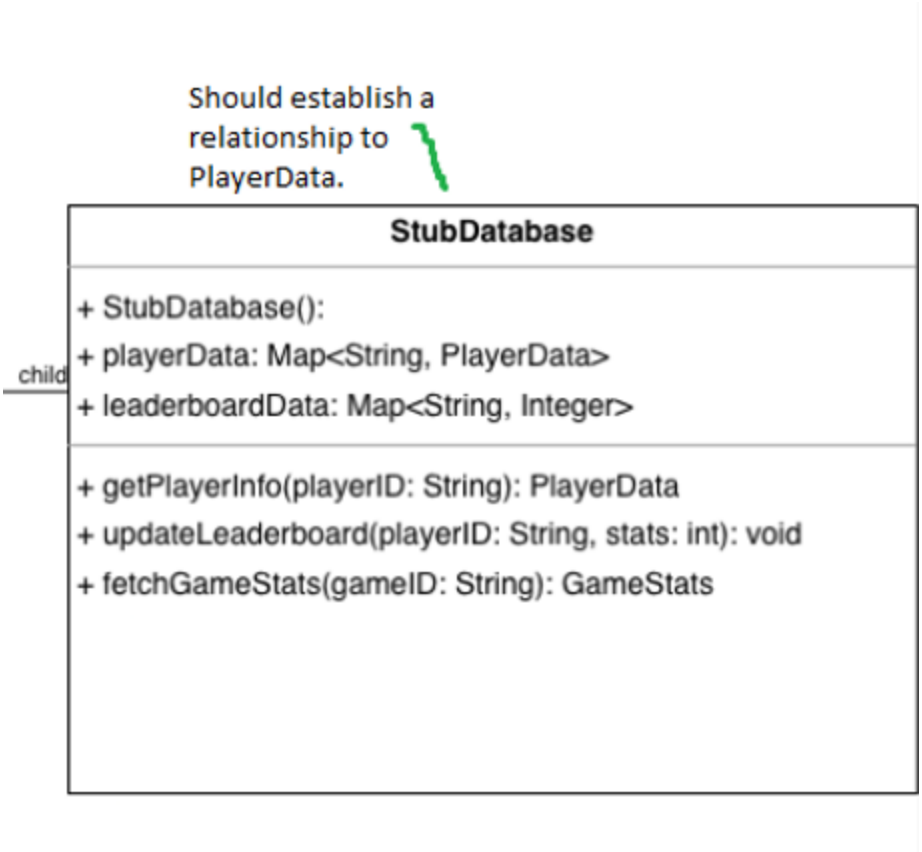
2.



a. **Add Arrows from *Connection* to *NetworkManager* to indicate it collects Connection type objects**

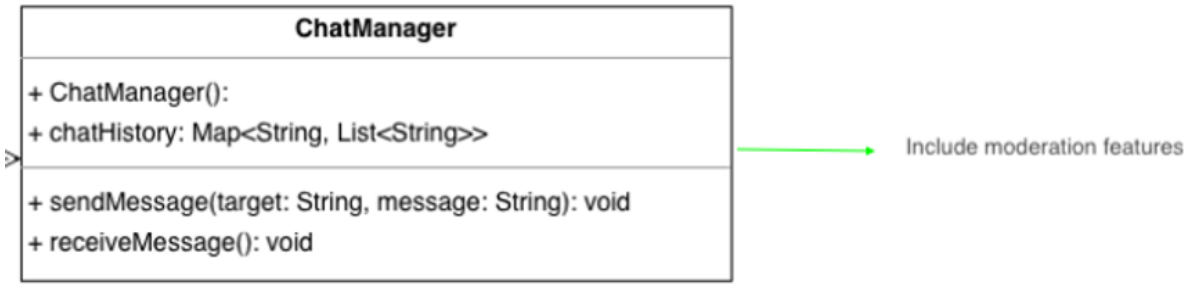
b. **Same thing for *Connection* to GameSessionManager**

3.



a. Make connection from *StubDatabase* to *PlayerData* in Class Diagram

4.

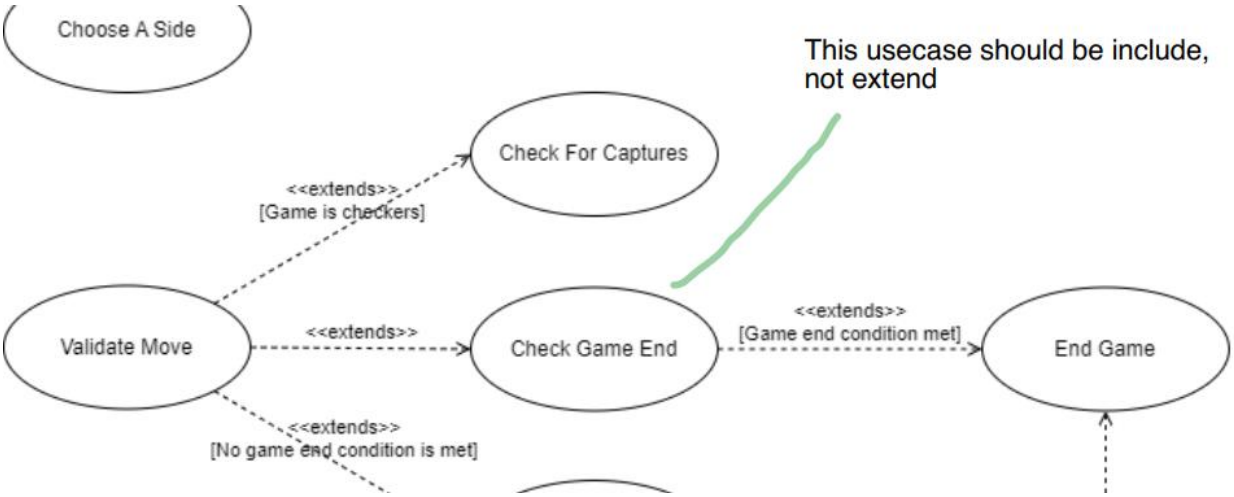


a. Team 1 proposed adding moderation features to ChatManager. Obviously not applicable.

USE-CASE DIAGRAMS:

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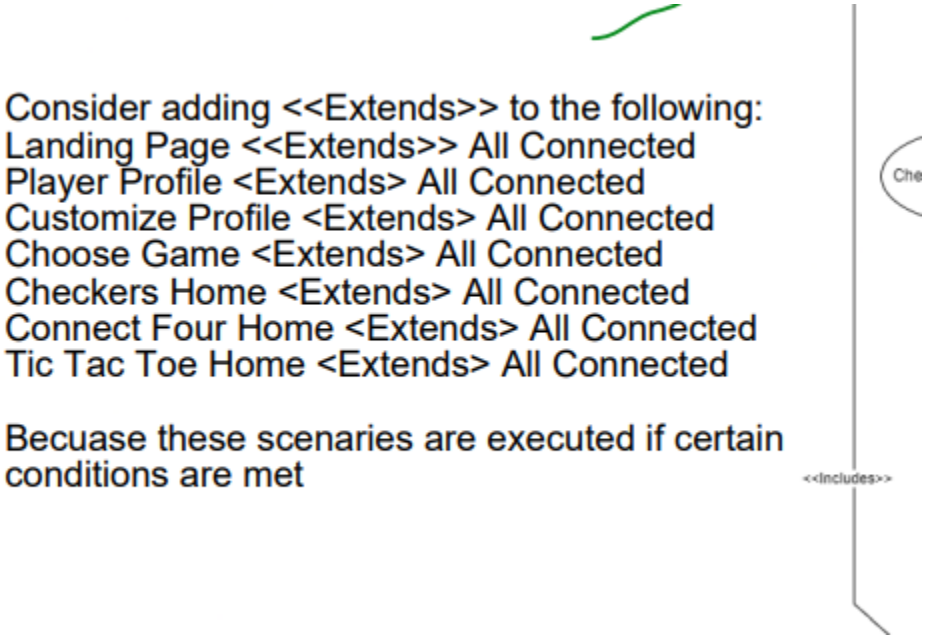
1.



a. **Change relationship from Validade Move to Check Game End to <<includes>>**

Frontend

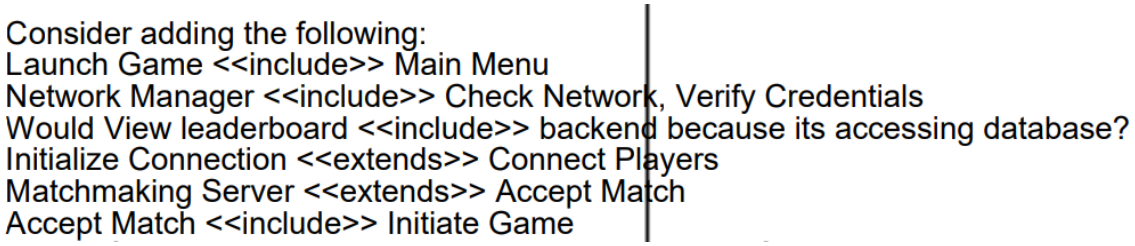
1.



a. **Self Explanatory:**
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